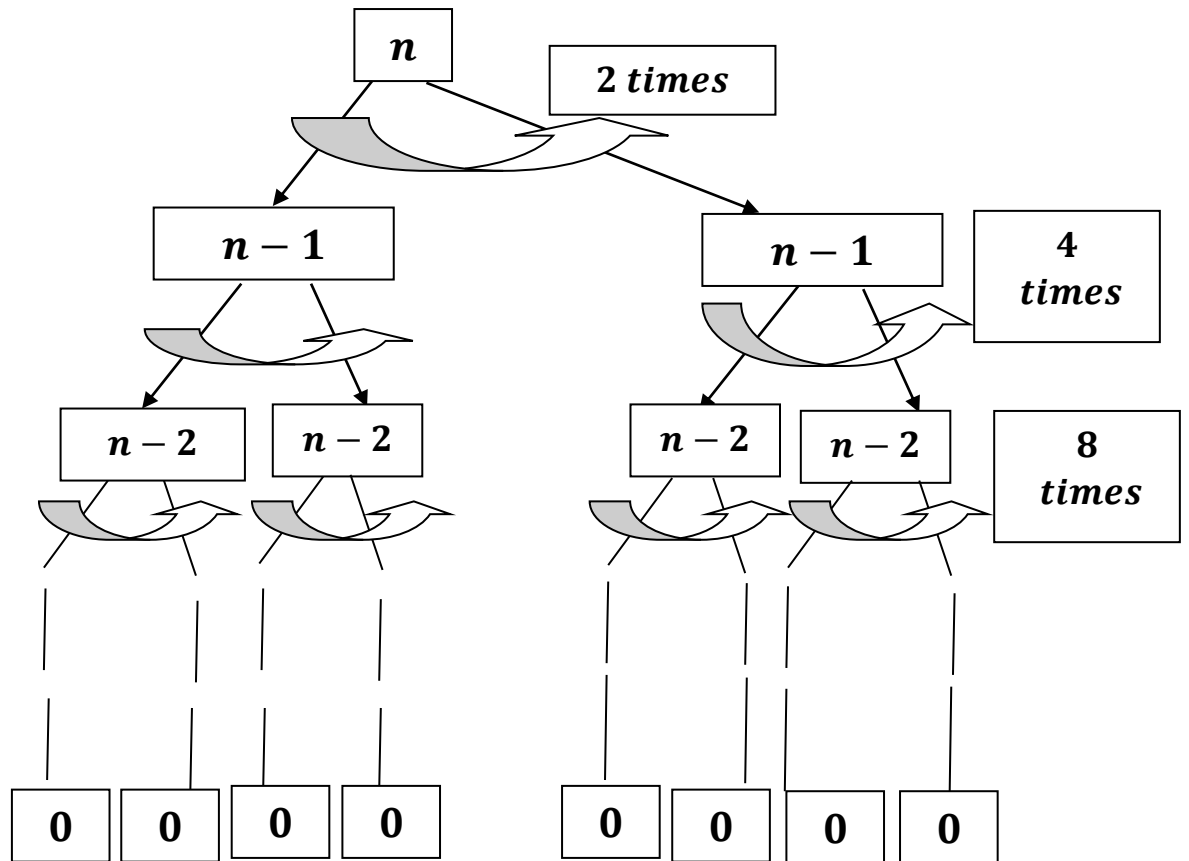


Tower Of Hanoi – Time Complexity calculation through Recursion Tree Method.



And recurrence relation given as follows:

$$T(n) = \begin{cases} 1 & , \text{if } n = 0 \\ 2T(n-1) + 1 & , \text{if } n > 0 \end{cases}$$

Note : we have discussed already earlier that when :

$$t_n = \begin{cases} 1 & \text{for } n = 1 \\ t_{n-1} + a & \text{for } n > 1 \end{cases}$$

When $a = 1$, problem size will remain constant i. e. 1,

Hence:

<i>Level</i>	<i>No. of problems</i>	<i>Problem size</i>	<i>Work Done</i>
0	1	1	1
1	2	1	2
2	4	1	4
.	.	.	.
.	.	.	.
$n - 1$	2^{n-1}	1	2^{n-1}

$$\Rightarrow 1 + 2 + 4 + \dots + 2^{n-1}$$

$$\Rightarrow 1 + 2 + 2^2 + \dots + 2^{n-1}$$

Applying geometric progression(series) :

$$S(n) = ar^0 + ar^1 + ar^2 + \dots + ar^n = a \left(\frac{r^{n+1} - 1}{r - 1} \right), \text{ where}$$

***S(n) is sum of first n terms , a is coefficient and
r is common ratio between the consecutive terms.***

We get:

$$\Rightarrow 1 \left(\frac{2^{n-1+1} - 1}{2 - 1} \right)$$

$$\Rightarrow 2^n - 1$$

putting Big O we get:

$$\Rightarrow O(2^n - 1)$$

$$\Rightarrow O(2^n) - O(1)$$

$$\Rightarrow O(2^n)$$
